

2.3 Arc Length & Change of Parameter

Recall: In calculus II you probably were introduced to the formula $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$ for the length of a

parametric curve in two dimensions. Since the distance formula in three dimensions is very closely related to the distance formula in two dimensions, this formula extends nicely in the 3-dimensional case. However, before we present this formula we introduce some terminology.

Definition: A parameterization $\mathbf{r}(t)$ is called smooth on an interval I if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$ for any t on I . A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Theorem: Suppose that $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ is a smooth vectored-value function for $a \leq t \leq b$, where the curve is only traversed one time as t goes from a to b then

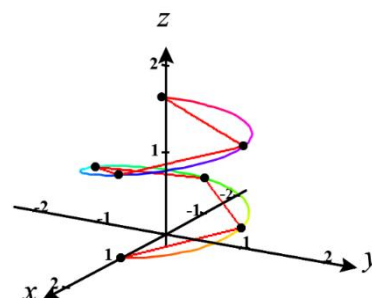
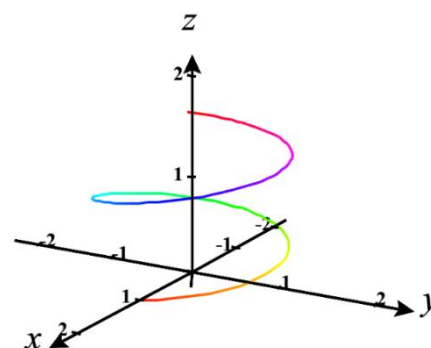
$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b \|\mathbf{r}'(t)\| dt .$$

Partial Proof:

Suppose we have a smooth vector-valued function given by $\mathbf{r}(t)$ where $a \leq t \leq b$.

We measure the length of any given line segment by creating a vector between the endpoints, which we will call $\mathbf{r}_i$, and calculating its magnitude.

Therefore, we estimate the total length of the vector function to be .



With a much more complicated argument it can be shown that the FTC and the MVT can be used to establish that

Therefore,

So, we have

Example 1: Find the length of the curve determined by $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j} + t\mathbf{k}$ from the point $(1, 0, 0)$ and $\left(-1, 0, \frac{\pi}{2}\right)$.

Definition: Suppose a curve C is given by $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ where $a \leq t \leq b$ where $\mathbf{r}'$ is continuous and C is traversed exactly once as t increases from a to b . We define the arc length function of C to be

$$s(t) = \int_a^t \|\mathbf{r}'(u)\| du \text{ for } t \text{ in } [a, b].$$

Note: Intuitively the arc length function of C takes in t and spits out the length of the part of C traversed from a to t .

Remark: By the fundamental theorem of calculus $\frac{ds}{dt} = \|\mathbf{r}'(t)\| \rightarrow ds = \|\mathbf{r}'(t)\| dt$. Therefore, as we will see

later in Chapter 5, the arc length integral is sometimes written as $L = \int_a^b ds$.

Example 2: Suppose C is given by $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j} + t\mathbf{k}$ where $0 \leq t \leq \frac{\pi}{2}$. Find the arc length function of C .

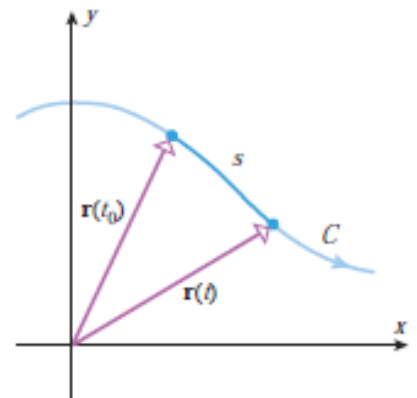
In the past several sections we have communicated curves in 3-space using parametric equations which used a parameter of t that we commonly thought of as representing time. However, it can be helpful to change parameters so that we can measure motion along a curve not just by some value of time. Specifically, it will be helpful to measure motion along a curve in terms of the distance we have travelled from some fixed starting point, that is to say, using arc length.

Definition: Suppose a curve C is given by $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ with $a \leq t \leq b$ where $\mathbf{r}$ is a smooth vector-valued function and C is traversed exactly once as t increases from a to b . If from the arc length function for C we can solve for t to obtain t as a function of s , then the arc length parameterization of C is $\mathbf{r}(t(s))$.

Steps to Obtaining an Arc Length Parameterization for a Curve:

1. Find the arc length function of C , $s(t)$.
2. Use $s(t)$ to solve for t in terms of s (this may not be possible). Then, you have t as a function of s .
3. Simply use composition of functions to obtain your arc length parameterization: $\mathbf{r}(t(s))$.

Remark: The power in the arc length parameterization is that if you plug a number like 2 into it, it will spit out a position vector whose terminal point is the point at which a particle which started its movement at the initial point of C is located after it has traveled 2 units along C .



Example 3: Suppose C is given by $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j} + t\mathbf{k}$ where $0 \leq t \leq \frac{\pi}{2}$.

a) Find an arc length parameterization of C .

b) Plug $s = \frac{\sqrt{5}\pi}{2}$ into the parameterization you found in (a). Look back in your notes to see why the result that you get is awesome.